

Lesson 07 · Chain and product rules

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

Use the chain rule to differentiate $y = (5x + 1)^3$.

☐ A $15(5x + 1)^2$

☐ B $3(5x + 1)^2$

☐ C $15(5x + 1)^3$

☐ D $5(5x + 1)^2$

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Q2 · Rank 1 / Foundation

Use the chain rule to differentiate $y = (x + 9)^5$.

☐ A $5x^4$

☐ B $5(x + 9)^4$

☐ C $9(x + 9)^4$

☐ D $5(x + 9)^5$

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Q3 · Rank 1 / Foundation

Use the chain rule to differentiate $y = (2x - 7)^8$.

☐ A $8(2x - 7)^7$

☐ B $16(2x - 7)^8$

☐ C $16(2x - 7)^7$

☐ D $-56(2x - 7)^7$

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Q4 · Rank 2 / Core

Use the chain rule to differentiate $y = (4x^2 - 7)^3$.

☐ A $3(4x^2 - 7)^2$

☐ B $12x(4x^2 - 7)^2$

☐ C $24x(4x^2 - 7)^3$

☐ D $24x(4x^2 - 7)^2$

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Q5 · Rank 2 / Core

Use the product rule to differentiate $y = x(3x + 4)^2$.

☐ A $(3x + 4)(9x + 4)$

☐ B $6x(3x + 4)$

☐ C $(3x + 4)^2$

☐ D $6(3x + 4)$

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Q6 · Rank 2 / Core

Use the product rule to differentiate $y = x^2(x - 2)^3$.

☐ A $6x(x - 2)^2$

☐ B $x(x - 2)^2(5x - 4)$

☐ C $2x(x - 2)^3$

☐ D $3x^2(x - 2)^2$

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Q7 · Rank 3 / Application

Differentiate $y = (5x + 7)^2$ by using the chain rule.

☐ **A** $10x + 14$

☐ **B** $50x + 14$

☐ **C** $50x + 70$

☐ **D** $25x + 70$

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Q8 · Rank 3 / Application

Differentiate $y = x^5(x + 2)^2$ by using the product rule.

☐ **A** $10x^4(x + 2)$

☐ **B** $5x^4(x + 2)^2$

☐ **C** $2x^5(x + 2)$

☐ **D** $x^4(x + 2)(7x + 10)$

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Q9 · Rank 3 / Application

Find an expression for marginal revenue when the demand function is $P = (100 - Q)^3$.

☐ A $MR = (100 - Q)^2(100 - 4Q)$

☐ B $MR = -3(100 - Q)^2$

☐ C $MR = (100 - Q)^3$

☐ D $MR = (100 - Q)^2(100 + 2Q)$

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Q10 · Rank 4 / Challenge · OPTIONAL

Use the quotient rule to differentiate $y = \frac{x}{x-5}$, where $x \neq 5$.

☐ A $\frac{5}{(x-5)^2}$

☐ B $-\frac{5}{(x-5)^2}$

☐ C 1

☐ D $\frac{1}{x-5}$

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